JEFF TARIMO

Head of Technology • IoT Systems Engineer • Full-Stack Cloud

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PROFESSIONAL SUMMARY

Multidisciplinary **Head of Technology & IoT Systems Engineer** with over 4 years of proven expertise designing, bench-testing, and deploying industrial-grade edge-to-cloud architectures. Specialized in technical leadership, microcontrollers (STM32, ESP32), low-power wireless networks (LoRaWAN, Cellular, BLE), and scalable backend pipelines (Python, Django, Node.js, AWS/Azure). Experienced in directing cross-functional engineering teams, architecting smart energy metering platforms, and delivering fault-tolerant telemetry infrastructure.

TECHNICAL EXPERTISE

Microcontrollers & Hardware: STM32, ESP32, Raspberry Pi, Arduino, Circuit Design, Sensor Integration

Firmware & Protocols: C/C++, MicroPython, MODBUS (RS485), MQTT, HTTP/S, UART, SPI, I2C

Wireless & Networking: LoRaWAN, Cellular (GPRS/LTE-M), Sigfox, BLE, Wi-Fi, CCNA R&S

Backend & Cloud: Node.js, Python (Django), Next.js, Redis, AWS, Azure, Linux (Ubuntu), REST APIs

Databases & Time-Series: ClickHouse, PostgreSQL, MySQL, InfluxDB (Time-series optimization)

Frontend & Leadership: React, Flutter, Tailwind CSS, HTML5, Node-RED Automation, Technical Strategy

WORK EXPERIENCE

Head of Technology — Powerpay Africa Limited Feb 2025 – Present | Nairobi, Kenya

- Direct overall technology vision, infrastructure scaling, strategic IT planning, and architectural roadmaps across hardware & software.
- Lead engineering teams across firmware, cloud backend, and mobile apps to deliver smart energy metering and financing systems.
- Orchestrate tech stack decisions, cloud operations (AWS/Azure/Ubuntu), and technical delivery to align with company growth goals.

Head of IoT — Powerpay Africa Limited Feb 2023 – Jan 2025 | Nairobi, Kenya

- Spearheaded firmware engineering using C++ and MicroPython for smart energy meters and edge telemetry controllers.
- Deployed multi-protocol communication pipelines (MODBUS RS485, GPRS, Wi-Fi, BLE) for continuous remote asset monitoring.
- Architected scalable backend services and REST APIs with Node.js, Django, ClickHouse, and PostgreSQL on Ubuntu cloud servers.
- Automated data normalization and exception alerts via custom Node-RED workflows, eliminating manual intervention.

Head of IoT — Upande Limited Sep 2022 – Jan 2023 | Nairobi, Kenya

- Directed product strategy and hardware delivery roadmaps as Product Owner across municipal and agricultural IoT initiatives.
- Streamlined device testing and dispatch pipelines using Node-RED, reducing commissioning time across field deployments.
- Liaised with enterprise stakeholders to translate complex monitoring requirements into scalable IoT architectures.

IoT Engineer — Upande Limited Apr 2021 – Aug 2022 | Nairobi, Kenya

- Assembled, calibrated, and deployed industrial sensor nodes (water level, ultrasonic, flow, weather) using LoRaWAN and GSM.
- Maintained field device fleets, diagnosing power faults and issuing remote firmware updates to maximize uptime.
- Configured InfluxDB time-series instances for sensor data ingestion, enabling real-time analytics and client dashboards.

Electrical Engineering Intern — New Kenya Co-operative Creameries Mar 2016 – Aug 2016 | Nyeri, Kenya

- Carried out scheduled electrical preventive maintenance across industrial dairy processing plants, reducing machine downtime.
- Assisted senior engineers in troubleshooting motor control panels, relay boards, and industrial automation equipment.

EDUCATION

M.Sc. in Computer Science
Zetech University
Ruiru, Kenya • Aug 2026 – Dec 2027

B.Sc. in Electronics & Computer Engineering
Jomo Kenyatta Univ. of Agriculture & Tech. (JKUAT)
Juja, Kiambu • Graduated 2021

CERTIFICATIONS & TRAINING

- **CCNA (Routing & Switching):** AFRALTI (2018)
- **Python & Web Development:** IST (2020)
- **Android App Development:** IST (2020)
- **Languages:** English (Fluent), Swahili (Fluent)